Causal Degradation-Guided Network With Spatial–Frequency Attention for Blind Hyperspectral Image Fusion

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***Abstract*—Fusing the spectral information of low-resolution hyperspectral image (LR-HSI) with the spatial details of high-resolution multispectral image (HR-MSI) is an effective strategy for reconstructing high-resolution hyperspectral image (HR-HSI). Existing methods perform well when the degradation characteristics of training and test data are consistent. However, their generalization ability often deteriorates when faced with unknown degradations. In this article, we treat the degradation conditions as confounders in the causal relationship, and introduce diverse degradation schemes to mitigate the spurious correlations of the fusion process, compelling the model to learn latent causal patterns independent of specific degradation mechanisms. For this, we propose a degradation-guided blind hyperspectral image fusion network with spatial-frequency attention to better reveal these underlying causal characteristics. Specifically, two U-shaped subnetworks adaptively extract multiscale spatial and spectral degradation features from LR-HSI and HR-MSI, which are then embedded as guidance features into the reconstruction network for more accurate HR-HSI reconstruction. In addition, we design a spatial–frequency attention module, which integrates spatial window self-attention and frequency-domain attention in a parallel and interactive manner to fully exploit the complementary characteristics of the spatial and frequency domains. Extensive experiments on multiple public datasets demonstrate the effectiveness and superior generalization of the proposed method. The code of the proposed approach is available on https://github.com/liuofficial/CDGN**

***Index Terms*—Blind fusion, causality, degradation guidance, frequency domain, hyperspectral image.**

I. Introduction

HYPERSPECTRAL image (HSI), consisting of dozens to hundreds of contiguous spectral bands, provides abundant spectral information. Such high spectral resolution enables HSI to be widely used in various fields including detection tasks [1], [2], image classification [3], [4], and object tracking [5]. However, due to the physical limitations of imaging systems, there is often a tradeoff between spatial and spectral resolutions. To overcome this limitation, fusing a low-resolution HSI (LR-HSI) with a high-resolution multispectral image (HR-MSI) has become an efficient and reliable approach to obtain a high-resolution HSI (HR-HSI), i.e., hyperspectral image fusion (HIF) [6], [7].

The development of HIF techniques can be roughly divided into two stages. Early studies mainly focused on exploring intrinsic structural priors of HSI [8], such as sparsity [9], low-rank property [10], [11], and total variation [12], [13]. These methods typically rely on explicit mathematical modeling and manually designed regularization terms, which offer strong theoretical interpretability and often exhibit robust generalization across a range of degradation patterns due to their model-based nature. Although they can improve reconstruction quality to some extent, they struggle to capture complex spatial–spectral relationships and are less robust to noise and unknown degradations. In addition, their optimization processes are often computationally expensive and lack generalization ability, which limits their performance in real-world scenarios.

In recent years, deep learning methods [14], [15], [16], [17], [18] have achieved remarkable progress by learning HSI priors from large-scale datasets, significantly advancing the field. These approaches typically construct end-to-end neural networks to directly establish complex mappings between input and output images, effectively capturing high-level features from input data and generating more accurate and detailed fusion images. Among these, convolutional neural networks (CNNs) [19], [20] have proven highly effective at capturing local spatial patterns and fine-grained textures of the input images. However, CNNs have relatively small receptive fields, which restricts their ability to capture global contextual features and thereby limits their overall performance. As an alternative to CNNs, Transformer [21] has demonstrated impressive performance by employing a self-attention mechanism that excels at modeling global interactions through token relationships.

Although CNN- and Transformer-based HIF methods have demonstrated strong capability in feature extraction and representation learning, most existing approaches are designed

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and trained under the assumption that the degradation process is fixed. Under this setting, models tend to learn appearance-driven mappings that are highly coupled with specific degradation patterns, while failing to adequately exploit the degradation prior information implicitly contained in the input images. Consequently, degradation-related features are prone to being entangled with the intrinsic image content during the fusion process, causing the model to develop a strong dependency on particular degradation conditions. When confronted with unseen, mixed, or more complex degradation scenarios, such degradation-specific mappings often fail to maintain robustness, ultimately leading to a significant decline in generalization performance.

To tackle the aforementioned challenges, we reconsider the HIF problem from the perspective of causal generation. From the generation mechanism, HSI reconstruction can be regarded as a causal generative process [22], in which an ideal HR-HSI undergoes a series of spatial and spectral degradation operations to produce the observed LR-HSI and HR-MSI. During this process, degradation factors such as spatial blurring and spectral sampling act as confounding variables that significantly influence the distribution of the observed data. Based on this, we propose a new training paradigm in which diverse training samples are generated by randomly adopting different point spread functions (PSFs) and spectral response functions (SRFs) from various sensors. By simulating multiple degradation conditions, this strategy effectively mitigates degradation-induced spurious correlations and encourages the network to learn latent causal patterns that are independent of specific degradation mechanisms, thereby improving robustness and generalization under unknown degradations.

Meanwhile, most existing HIF methods primarily focus on recovering spatial details and spectral information from LR-HSI and HR-MSI, the degradation process is typically treated as a fixed or implicit prior, and the degradation information embedded in the input data is not sufficiently exploited. As a result, the learned reconstruction remains entangled with degradation-related information and fails to fully reflect the underlying causal mechanism. Based on this observation, to better exploit degradation characteristics and explore the latent causal principles in the reconstruction process, we employ two U-Net [23] subnetworks to estimate the residual maps between the LR-HSI and HR-MSI inputs and their corresponding reference images, respectively. These residual maps characterize the spatial and spectral degradation features of the input data. Subsequently, the degradation features extracted from the decoder sides of the subnetworks are embedded into the reconstruction network through a dual-branch spatial–spectral degradation guidance module (DGM), enabling more accurate high-resolution HSI reconstruction. Furthermore, we design a spatial–frequency attention module (SFAM) that jointly and interactively integrates spatial window self-attention and frequency-domain attention in parallel, fully exploiting the complementary information between the spatial and frequency domains to enhance spatial detail reconstruction and spectral consistency.

The main contributions of this work are summarized as follows.

1) We revisit the HIF task from a causal inference perspective, where training samples are generated by randomly adopting diverse PSFs and SRFs to simulate various degradation conditions, thereby encouraging the model to learn latent causal patterns that are independent of specific degradation mechanisms.
2) To further leverage degradation characteristics for discovering causal patterns, we construct a degradation-guided network with spatial–frequency attention. This network guides the model to learn degradation-invariant causal representations, thereby the robustness and generalizability of the reconstruction.
3) We design the DGM that injects decoder-side intermediate features from the two U-shaped subnetworks into the reconstruction network. These features contain spatial–spectral degradation information and improve HR-HSI reconstruction accuracy.
4) To fully exploit the complementary properties of spatial and frequency domains, we design the SFAM, which parallelly and interactively integrates spatial window attention with frequency-domain attention to jointly enhance spatial detail and spectral consistency.

The rest of this article is organized as follows. Section II reviews the current research on HIF. Section III introduces the proposed method, network architecture, and its components. Section IV presents and discusses the experimental results on multiple datasets. Finally, Section V concludes this article.

II. Related Work

A. Traditional Methods

Traditional HIF methods based on the fusion of LR-HSI and HR-MSI can be broadly categorized into three classes: Bayesian-based methods, matrix factorization-based methods, and tensor factorization-based methods.

Bayesian-based methods [11], [24], [25], [26] provide a probabilistic modeling framework for image fusion. They establish a generative relationship between the observed images (LR-HSI and HR-MSI) and the latent HR-HSI, and estimate the posterior distribution via Bayesian inference. This allows the fusion process to simultaneously account for data consistency and prior knowledge constraints. Early studies, such as Eismann and Hardie [24] built a Bayesian estimation framework based on the stochastic mixing model, laying a theoretical foundation for subsequent HSI enhancement and fusion. Wei et al. [25] combined the Bayesian framework with numerical optimization, solving the Sylvester equation and employing the alternating direction method of multipliers to achieve an efficient fusion process. Akhtar et al. [26] proposed a Bayesian sparse coding method for HIF, using the beta process to learn dictionaries and adaptively model the sparse structure of HSI.

Matrix factorization-based methods [9], [13], [27], [28] model HSI as a linear combination of endmember spectra and abundance matrices, thereby enabling joint modeling of spectral and spatial information. In early work, Yokoya et al. [27] proposed the coupled nonnegative matrix factorization (CNMF) method, which simultaneously factorizes

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the HSI and MSI under nonnegativity constraints, capturing complementary spectral and spatial resolution information from both modalities. Subsequent research has extended the matrix factorization framework. For example, Dong et al. [9] proposed a nonnegative matrix Factorization for HIF, which jointly learns a spectral dictionary from the low-resolution input and estimates structured sparse codes using clustering to exploit spatial–spectral correlations. Sim ões et al. [13] introduced a low-dimensional subspace assumption and vector total variation regularization within the matrix factorization framework. By constructing a convex optimization model, this approach preserves spectral consistency while enhancing spatial edge features.

Tensor factorization-based methods [10], [29], [30], [31] directly model 3-D or even higher-order image tensors, enabling more effective feature extraction and information fusion without destroying the intrinsic multidimensional structure of the data. For example, Kanatsoulis et al. [29] exploited the intrinsic tensor structure of LR-HSI and HR-MSI to achieve identifiable HR-HSI recovery without requiring accurate knowledge of degradation operators. Subsequently, Li et al. [30] introduced the Coupled Sparse Tensor Factorization method, which incorporates sparsity constraints and nonlocal priors during tensor decomposition, effectively enhancing reconstruction performance in complex textured regions. To further improve fusion robustness, Dian et al. [10] proposed a low-rank hyperspectral fusion model based on the low tensor-train rank concept. Recent studies have additionally integrated Bayesian inference and nonlocal patch modeling, such as Bayesian nonlocal patch tensor factorization [31], achieving a better balance between model rank estimation and feature representation capability.

B. Deep Learning-Based Methods

CNNs have demonstrated remarkable capability in HSI processing owing to their local receptive fields, weight sharing, and hierarchical feature learning mechanisms. Compared with traditional handcrafted features or matrix factorization–based methods, CNNs can automatically learn spatial–spectral representations, enabling more effective capture of image details and spectral structures. In HIF tasks [32], [33], [34], [35], [36], [37], [38], [39], CNNs are able to model complex spatial–spectral relationships and exploit multiscale contextual information through deep convolutional architectures, thereby achieving high-quality image reconstruction. Early approaches, such as that of Palsson et al. [32] employed 3-D convolutions to simultaneously process spatial and spectral dimensions while incorporating dimensionality reduction to alleviate computational costs. Dian et al. [33] further integrated denoisers or regularization constraints into optimization frameworks, embedding CNNs to enhance fusion stability and spectral fidelity. Hu et al. [37] combined attention mechanisms with pixel-shuffle modules to improve spatial–spectral feature recovery. In subsequent studies, to better exploit multiscale information, Ran et al. [38] introduced a high-resolution-guided CNN framework that compensates for low-resolution image details between multiscale guidance and reconstruction branches while maintaining spectral consistency. Wang et al. [39] combined multiscale convolutions with axial attention and proposed a progressive feature interaction and fusion strategy to fully exploit both spatial and spectral information.

Although CNNs perform well in local feature extraction, their limited receptive fields hinder capturing global information. Optimized visual transformers [40], [41], [42], [43], [44], [45] provide efficient modeling of global relationships and have found successful application in HIF. For example, Hu et al. [40] used a dual-branch self-attention module to simultaneously model spatial and spectral features, improving global image consistency. Sun et al. [42] proposed a multiscale deep cross-fusion Transformer network that achieves progressive reconstruction through multiscale spatial–spectral feature interactions. Fang et al. [43] introduced a multi-input multioutput spatial–spectral Transformer network that enhances reconstruction of image edges and structural details via multihead feature attention, a multiscale convolutional gated feedforward network, and wavelet-based high-frequency loss. Ma et al. [45] proposed a bidirectional cross Transformer that effectively strengthens mutual information interaction and self-attention modeling between hyperspectral and multispectral images through directional multihead cross attention and Swin Transformer blocks. Mamba-based state space models also possess the capability to model long-range dependencies while avoiding the quadratic computational complexity inherent in Transformers, and have attracted increasing attention in HIF tasks. For example, Zhu et al. [46] proposed the Mamba cooperative implicit neural representation fusion network that achieves high-quality fusion through global continuous feature modeling with lightweight computation. Zhao et al. [47] presented a CNN–Mamba fusion network that leverages mutual guidance between hyperspectral and multispectral images and combines CNN with Mamba to effectively capture local and global features.

However, CNNs and Transformers largely focus on spatial information, neglecting frequency-domain features and global structures. Combining spatial and frequency-domain approaches [48], [49], [50], [51] can enhance super-resolution performance; thus, Fourier transforms have been introduced into HIF to provide a global perspective with long-range dependencies. For example, Guo et al. [48] integrated spatial- and frequency-domain modeling through a multiscale encoder–decoder architecture and an across-domain consistency difference module to capture high-frequency details, while using dilated convolutions to flexibly adjust the receptive field. Xu et al. [51] proposed an enhanced spatial-frequency collaborative mechanism that jointly models spatial and frequency features, thereby balancing global contextual information and local detail reconstruction. Furthermore, to better exploit global structures, high-frequency cues, and the long-range dependency modeling capability of self-attention, several studies [52], [53] have combined frequency-domain representations with Transformer architectures to achieve joint spatial–frequency modeling and fine-grained detail enhancement for HSI reconstruction. In contrast to prior frequency-domain methods that rely on independent or loosely coupled spatial–frequency modeling, our

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approach enables deeper integration by establishing bidirectional feature interactions between the spatial and frequency domains.

In addition to these architectural categories, generative models have emerged as a powerful paradigm for HIF. Generative adversarial network (GAN)-based methods [54], [55] employ adversarial training to enhance perceptual quality and spatial details, though they may suffer from training instability. For instance, Liu et al. [54] proposed R2D2-GAN, a robust dual discriminator GAN for microscopy HSI super-resolution, which leverages dual discriminators to improve reconstruction fidelity. Shi et al. [55] introduced a latent encoder coupled GAN that efficiently maps low-resolution HSI to high-resolution counterparts while preserving spectral consistency. Denoising diffusion probabilistic model-based methods [56], [57] have also shown remarkable potential in HIF. Dong et al. [56] developed ISPDiff, an interpretable scale-propelled diffusion model that incorporates scale-aware constraints to guide the diffusion process for high-quality HSI reconstruction. Zhu et al. [57] proposed a self-learning adaptive residual guided subspace diffusion model that only utilizes observed images, using two lightweight diffusion models to separately learn spectral and spatial distributions and reconstruct HR-HSI. These methods progressively refine random noise into high-quality HR-HSIs, achieving remarkable realism but at the cost of high inference latency.

Despite these advances, existing methods often struggle with generalization under unknown degradation patterns, as they primarily learn degradation-specific mappings from training data. This limitation motivates our causal degradation-guided approach presented in this article.

C. Causal Inference

Causal reasoning [22] aims to uncover the causal relationships between variables and eliminate biases introduced by confounding factors. In deep learning, traditional methods often focus on fitting correlations between inputs and outputs while ignoring underlying causal structures, making them prone to capturing spurious correlations. To address this issue, several studies [58], [59] have introduced causal inference into visual recognition tasks. Among them, Lu et al. [58] proposed a causally inspired representation learning (CIRL) framework, which formulates domain generalization using structural causal models and extracts label-related yet domain-invariant causal factors from input data. Building upon CIRL, Xu and Shao [59] further investigated the overlooked problem of feature dependence in domain generalization, noting that dependent features may induce spurious causal associations. To address this, they employed GAN-based models to explicitly learn independent causal representations.

In the context of HSI analysis, causal inference has been introduced into classification tasks. Specifically, Cheng et al. [60] proposed a causal meta-transfer learning approach that separates causal and noncausal factors to alleviate sample selection bias and enhance cross-domain few-shot classification performance. Similarly, Dong et al. [61] developed a causal constraint network for hyperspectral cross-scene classification, which reduces source–target domain discrepancies through data enhancement and causal alignment, thereby improving single-domain generalization. Furthermore, Wang et al. [62] introduced a causal meta-generalization framework that induces domain-invariant causal meta-knowledge, enabling robust classification on unseen target domains.

In this work, to the best of our knowledge, we make the first attempt to introduce a causal perspective into HIF. We treat the degradation conditions in HIF as confounding factors and reduce their influence, thereby forcing the model to learn latent causal relationships that are independent of degradation. This enhances the generalization capability of the reconstruction network under various spatial and spectral degradations.

III. Methodology

In this section, we begin by revisiting the HIF task from a causal perspective. Then, we introduce the proposed blind fusion network, followed by the implementation details of each module within the architecture and the loss functions employed.

For simplicity, we denote the LR-HSI as $\mathbf{Y} \in \mathbb{R}^{C\times hw}$, and the HR-MSI as $\mathbf{Z} \in \mathbb{R}^{c\times HW}$. The desired HR-HSI is represented as $\mathbf{X} \in \mathbb{R}^{C\times HW}$, which contains both rich spatial and spectral information. Here, C/c denote the number of spectral bands, H/h and W/w represent the spatial resolutions. Due to inevitable physical imaging limitations, they usually satisfy $C \gg c$, $H \gg h$ and $W \gg w$. $\mathbf{Y}$ and $\mathbf{Z}$ are commonly treated as observations derived from $\mathbf{X}$ after undergoing spatial and spectral degradations, respectively. More precisely, the relationship between the observed images and the target output can be formulated as follows:

$$\mathbf{Y} = \mathbf{XBH} + \mathbf{N}_y, \quad \mathbf{Z} = \mathbf{RX} + \mathbf{N}_z \tag{1}$$

where $\mathbf{R} \in \mathbb{R}^{c\times C}$ denotes the SRF, $\mathbf{B} \in \mathbb{R}^{HW\times HW}$ denotes the spatial blur operator, and $\mathbf{H} \in \mathbb{R}^{HW\times hw}$ represents the spatial downsampling operator. $\mathbf{N}_y$ and $\mathbf{N}_z$ represent sensor noise or modeling errors in the LR-HSI and HR-MSI, respectively.

A. Causal Modeling of HIF

HIF aims to reconstruct HR-HSI $\mathbf{X}$ from its corresponding LR-HSI $\mathbf{Y}$ and HR-MSI $\mathbf{Z}$. The degradation process can be formulated as

$$\mathbf{Y}, \mathbf{Z} = g(\mathbf{X}, \mathbf{D}), \quad \mathbf{D} = \mathbf{R}, \mathbf{B}, \mathbf{H}, \mathbf{N}_y, \mathbf{N}_z. \tag{2}$$

Here, g represents the spatial and spectral degradation processes. $\mathbf{D}$ denotes the set of all possible degradation conditions.

The HIF network f is trained via a loss function to minimize the discrepancy between the reconstructed HSI $\mathbf{X}_r$ and the ground-truth image $\mathbf{X}$. We model this process using a Structural Causal Graph, as illustrated in Fig. 1(a). In this graph, $\mathbf{D} \rightarrow (\mathbf{Y}, \mathbf{Z}) \leftarrow \mathbf{X}$ represents the degradation process, $(\mathbf{Y}, \mathbf{Z}) \rightarrow \mathbf{X}_r$ represents the reconstruction process performed by the network f, and $\mathbf{D} \rightarrow \mathbf{X}_r$ denotes the influence of degradation information on the reconstructed image $\mathbf{X}_r$. From a causal perspective, learning causal representations requires the network f to capture the causal effect $(\mathbf{Y}, \mathbf{Z}) \rightarrow \mathbf{X}_r$.

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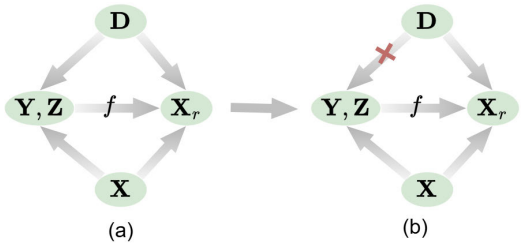


Fig. 1. Structural causal graph. (a) Structural causal graph of HIF. (b) Backdoor adjustment.

However, there exists a confounding path $(\mathbf{Y},\mathbf{Z}) \leftarrow \mathbf{D} \rightarrow \mathbf{X}_r$. The network f learns the $(\mathbf{Y},\mathbf{Z}) \rightarrow \mathbf{X}_r$ mapping from training data containing $\mathbf{D}$-induced correlations, making $\mathbf{D}$ a confounder that introduces spurious input-target correlations. The confounder $\mathbf{D}$ limits the generalization capability of the network because the conditional probability learned by the network, $P(\mathbf{X}_r \mid \mathbf{Y},\mathbf{Z})$, actually depends on the specific degradation condition $d \in \mathbf{D}$, i.e., $P(\mathbf{X}_r|\mathbf{Y},\mathbf{Z},d)$. Consequently, when encountering unseen degradation conditions, the network may fail to generalize effectively.

To address this confounding issue, we draw upon causal inference theory, specifically the backdoor adjustment criterion, as illustrated in Fig. 1(b). Then, the causal effect of observations $(\mathbf{Y},\mathbf{Z})$ on the reconstruction $\mathbf{X}_r$ can be estimated by marginalizing over the confounder $\mathbf{D}$

$$P\left(\mathbf{X}_r|do\left(\mathbf{Y},\mathbf{Z}\right)\right) = \sum_{d\in\mathbf{D}} P\left(\mathbf{X}_r|\mathbf{Y},\mathbf{Z},d\right) P\left(d\right) \tag{3}$$

where "do" operator blocks the confounding path from $\mathbf{D}$ to $(\mathbf{Y},\mathbf{Z})$. It states that if we can intervene on the confounder $\mathbf{D}$ and make it independent of $\mathbf{X}$, we can block the confounding path and recover the true causal effect.

To simplify the optimization of (3), we assign an equal probability of $1/|\mathbf{D}|$ to each distortion d, with $|\mathbf{D}|$ representing the total count of distortion types and levels. Consequently, we adopt a multidegradation training strategy to implement the backdoor adjustment. Instead of training on a fixed degradation process, we construct a diverse set of degradation conditions $\mathbf{D}$ (detailed in Section IV-A). By randomly sampling degradation $d \in \mathbf{D}$ during training, we simulate the intervention on $\mathbf{D}$, effectively making $\mathbf{D}$ independent of $\mathbf{X}$. This process approximates the summation in (3), compelling the network to learn the invariant causal mechanism $(\mathbf{Y},\mathbf{Z}) \rightarrow \mathbf{X}_r$, that is independent of specific degradation artifacts. The optimization objective is formulated as

$$\theta^* = \arg\min_{\theta} \mathbb{E}_{(\mathbf{Y}_d,\mathbf{Z}_d,\mathbf{X})\sim\mathcal{D}} \left[\sum_{d\in\mathbf{D}} \mathcal{L}\left(f_\theta\left(\mathbf{Y}_d,\mathbf{Z}_d\right),\mathbf{X}\right)\right]. \tag{4}$$

Here, $\mathbf{Y}_d$ and $\mathbf{Z}_d$ represent the training data generated under a specific degradation condition $d \in \mathbf{D}$, $\mathcal{D} = \{\mathbf{Y}_d,\mathbf{Z}_d,\mathbf{X}|d \in \mathbf{D}\}$ denotes the entire training dataset, $\mathcal{L}$ is the loss function defined in Section III-F and θ represents the network parameters. The formulation (4) ensures that the learned representation is invariant to the specific degradation conditions, thereby enhancing generalization to unseen real-world scenarios.

Ideally, we aim to construct a sufficiently rich set of degradation conditions $\mathcal{D}$ such that unseen real-world degradations fall within its distribution. However, this is practically infeasible due to the complexity of real-world scenarios. To achieve the decoupling of the network f from specific degradation conditions $\mathcal{D}$, we further address this challenge from the perspective of network architecture by designing two degradation-aware U-Nets (detailed in Section III-C). Ultimately, the proposed multidegradation training strategy and the degradation-aware U-Nets synergistically to realize robust decoupling.

B. Overall Network Architecture

Our proposed HIF network, as illustrated in Fig. 2(a), mainly consists of four modules: the U-Net network $\boldsymbol{U}_{\text{spa}}$ that extracts spatial degradation information from the LR-HSI; the U-Net network $\boldsymbol{U}_{\text{spe}}$ used to extract spectral degradation information from the HR-MSI; the SFAM, which fully utilizes spatial and frequency domain information for image restoration; and the DGM, which embeds the extracted spatial and spectral degradation information into the reconstruction network.

First, to match the spatial sizes of the two input images, bilinear interpolation is applied to upsample $\mathbf{Y}$ to the same spatial sizes as $\mathbf{Z}$, obtaining $\mathbf{Y}_{up}$. Then, a 1×1 convolution is used to increase the channel number of $\mathbf{Z}$ to match that of $\mathbf{Y}$, resulting in $\mathbf{Z}_{up}$. Next, we concatenate $\mathbf{Y}_{up}$ and $\mathbf{Z}_{up}$ along the channel dimension and perform an initial fusion using a 1×1 convolution. Meanwhile, we use $\boldsymbol{U}_{\text{spa}}/\boldsymbol{U}_{\text{spe}}$ to learn the residuals between $\mathbf{Y}_{up}/\mathbf{Z}_{up}$ and the reference image, while extracting multiscale features from the encoder sides for subsequent reconstruction assistance. To align the multiscale intermediate information extracted by the U-shaped networks, we formulate the reconstruction as a multilevel procedure and perform corresponding hierarchical operations. Between each level, average pooling with a kernel size of 2 and a stride of 2 is applied for downsampling. At each level, the SFAM is first used to preliminarily restore information in both spatial and frequency domains, and then the DGM integrates the intermediate features extracted from $\boldsymbol{U}_{\text{spa}}$ and $\boldsymbol{U}_{\text{spe}}$, which contain rich spatial and spectral degradation information, to further assist the reconstruction process. Finally, the restored features from all levels are upsampled to align their spatial dimensions and fused using a 1×1convolution. By exploiting intermediate features that encode rich degradation cues as auxiliary guidance, the proposed network facilitates more accurate and higher-fidelity reconstruction of HR-HSI.

C. Degradation-Aware U-Nets

In HIF, a key challenge lies in recovering high-dimensional features distorted by spatial–spectral degradation. In contrast to conventional approaches that focus exclusively on reconstructing high-resolution details, our method exploits the structured prior inherently embedded in the degradation process—specifically, the spatial and spectral blurring patterns—which serves as a valuable constraint to guide the

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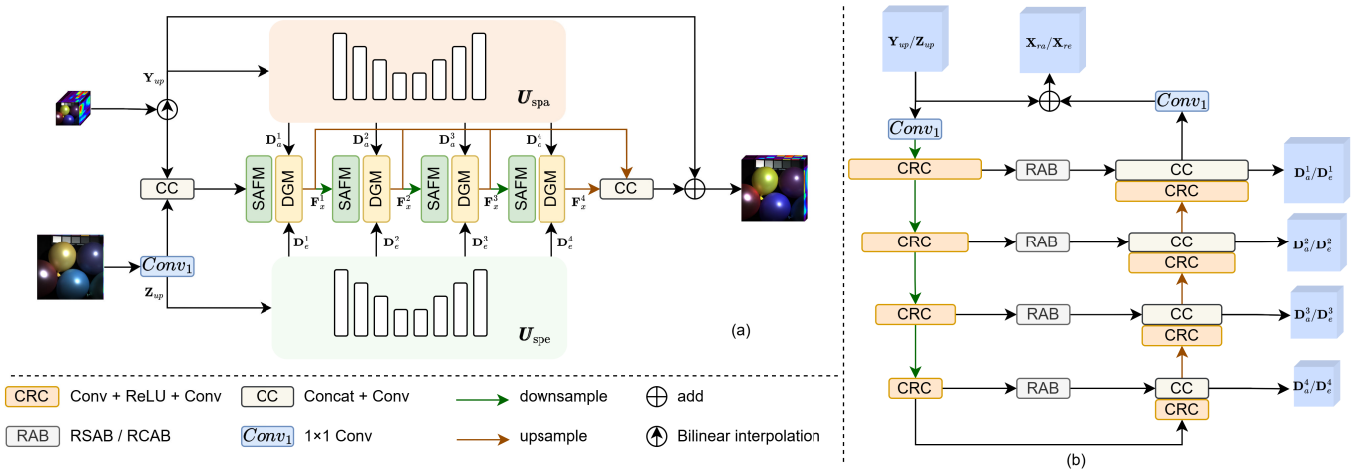


Fig. 2. Architecture of the proposed network. (a) Overview of network. (b) Structures of spatial U-Net ($\boldsymbol{U}_{\text{spa}}$) and spectral U-Net ($\boldsymbol{U}_{\text{spe}}$) for degradation feature extraction.

reconstruction. To this end, we design two dedicated U-Net networks for extracting degradation-aware features: one for spatial degradation ($\boldsymbol{U}_{\text{spa}}$) and one for spectral degradation ($\boldsymbol{U}_{\text{spe}}$). It should be noted that although the inputs to these networks, the up-sampled LR-HSI $\mathbf{Y}_{up}$ and HR-MSI $\mathbf{Z}_{up}$, are obtained through simple interpolation and channel transformation rather than genuine high-resolution References, they preserve the essential spectral distributions and spatial structure skeletons of the original low-resolution observations. Compared to the ground truth high-resolution image $\mathbf{X}$, $\mathbf{Y}_{up}$ primarily lacks high-frequency spatial details (manifested as blurriness), while $\mathbf{Z}_{up}$ primarily suffers from spectral discontinuity (manifested as spectral distortion). Different degradation mechanisms (e.g., PSF and SRF) leave distinct identifiable patterns in these interpolated images. By training on large-scale data with diverse degradations, the U-Nets learn to map these observable patterns to the corresponding degradation residuals, rather than attempting direct reconstruction of the high-resolution image. Both networks are end-to-end trainable (see the loss function in Section III-F) and yield two estimations of the reference image, thereby compelling the networks to learn the degradation residual maps between the input images and the reference image.

1) $\boldsymbol{U}_{Spa}$ Network: The spatial U-Net $\boldsymbol{U}_{\text{spa}}$ is used to extract spatial degradation-aware features, which produces the spatial degradation residual map $\mathbf{D}_{spa}$ and the spatially enhanced result $\mathbf{X}_{ra}$

$$\mathbf{X}_{ra} = \mathbf{D}_{spa} + \mathbf{Y}_{up} = \boldsymbol{U}_{\text{spa}}\left(\mathbf{Y}_{up}\right) + \mathbf{Y}_{up}. \tag{5}$$

In the encoder, each layer consists of two 3×3 convolutions saparated by a ReLU activation function. Downsampling between layers is achieved through average pooling with a of 2×2 kernel and a stride of 2. In the decoder, a structure similar to the encoder is used, along with a transposed convolution with a 3×3 kernel and a stride of 2 for upsampling. Additionally, we introduce attention mechanisms for $\boldsymbol{U}_{\text{spa}}$. Specifically, a residual SA block (RSAB) is incorporated into $\boldsymbol{U}_{\text{spa}}$, which takes encoder features as input and selectively highlights the most informative cues along the spatial dimension to better capture degradation patterns. The operations can be formulated as

$$\begin{aligned}\mathbf{F}_a' &= \text{RSAB}\left(\mathbf{F}_a\right) \\ &= \mathbf{F}_a + \mathbf{F}_a \cdot \sigma\left(conv_1\left(cat\left(mean\left(\mathbf{F}_a\right), max\left(\mathbf{F}_a\right)\right)\right)\right).\end{aligned} \tag{6}$$

Here, $\mathbf{F}_a$ denote shallow features from the encoder sides of $\boldsymbol{U}_{\text{spa}}$. $conv_1$ denotes a 1×1 convolution, *cat* indicates concatenation along the channel dimension, σ denotes the sigmoid function, *mean* and *max* denote channel-wise average and max pooling. The refined features are then fused with decoder features using a 1×1 convolution to obtain multiscale spatial degradation representations $\mathbf{D}_a^i$ $(i = 1, 2, 3, 4)$.

2) $\boldsymbol{U}_{Spe}$ Network: To characterize spectral degradation such as band mixing and SRF-induced spectral blurring, we design the spectral U-Net $\boldsymbol{U}_{\text{spe}}$, which generates the spectral degradation residual map $\mathbf{D}_{spe}$ and the spectrally enhanced output $\mathbf{X}_{re}$

$$\mathbf{X}_{re} = \mathbf{D}_{spe} + \mathbf{Z}_{up} = \boldsymbol{U}_{\text{spe}}\left(\mathbf{Z}_{up}\right) + \mathbf{Z}_{up}. \tag{7}$$

The encoder–decoder structure follows the same design as $\boldsymbol{U}_{\text{spa}}$. To emphasize informative spectral cues, we incorporate a residual CA block (RCAB) into $\boldsymbol{U}_{\text{spe}}$

$$\begin{aligned}\mathbf{F}_e' &= \text{RCAB}\left(\mathbf{F}_e\right) \\ &= \mathbf{F}_e + \mathbf{F}_e \cdot \sigma\left(LRL\left(GAP\left(\mathbf{F}_e\right)\right) + LRL\left(GMP\left(\mathbf{F}_e\right)\right)\right).\end{aligned} \tag{8}$$

Here, $\mathbf{F}_e$ denote shallow features from the encoder sides of $\boldsymbol{U}_{\text{spe}}$. *LRL* consists of two linear layers connected by a ReLU activation function, and *GAP* and *GMP* denote global average pooling and global max pooling, respectively. These features are fused with decoder-side features to produce multiscale spectral degradation representations $\mathbf{D}_e^i$ $(i = 1, 2, 3, 4)$.

After extracting the multiscale spatial and spectral degradation feature sets $\mathbf{D}_a$ and $\mathbf{D}_e$, they are integrated into the subsequent reconstruction network to guide the recovery of HR-HSI. Fig. 2(b) provides a visual illustration of the two networks.

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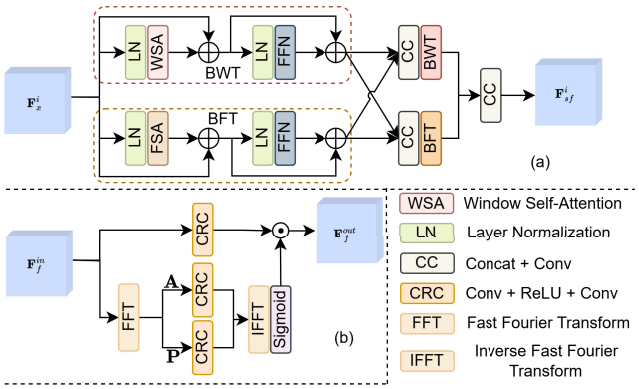


Fig. 3. Framework of the proposed SFAM. Here, "**A**" denotes amplitude, "**P**" denotes phase. (a) Overview of SFAM. (b) Structure of FSA.

D. Spatial-Frequency Attention Module

The frequency-domain information is beneficial for reconstruction. Unlike the spatial domain, which mixes structure and intensity, the frequency domain provides a disentangled representation by decomposing information into amplitude and phase, with the former governing texture contrast and spectral energy, and the latter encoding spatial structure and edge locations. This decomposition is particularly valuable for HIF because spatial and spectral degradations affect amplitude and phase differently, making frequency-domain modeling essential for disentangling degradation effects from intrinsic image content. Sole reliance on spatial-domain learning constrains the model's capacity to perceive both global structures and high-frequency details, often resulting in missing high-frequency details or degraded spectral consistency in the reconstructed images. To address this limitation, we propose the SFAM, which synergistically integrates window-based attention Transformers with frequency-domain Transformers, as illustrated in Fig. 3.

The main idea of SFAM is to enable bidirectional interaction and fusion between spatial and frequency domain features, in order to fully exploit their complementary information. The input $\mathbf{F}_x^i$ is fed into two branches simultaneously: one is a spatial branch based on windowed attention Transformer (BWT), and the other is a frequency branch based on frequency-domain Transformer (BFT). After extracting features in the spatial and frequency domains, respectively, the two branches interact through cross-connections: spatial features guide the extraction of frequency-domain information, while frequency-domain features in turn enhance the modeling of spatial structures. Finally, the fused features from both branches are integrated via concatenation to form the output $\mathbf{F}_{sf}^i$. The operations can be formulated as follows:

$$\begin{aligned}
\mathbf{F}_s &= \mathrm{BWT}\left(\mathbf{F}_x^i\right) \\
\mathbf{F}_f &= \mathrm{BFT}\left(\mathbf{F}_x^i\right) \\
\mathbf{F}_{sf} &= \mathrm{BWT}\left(conv_1\left(cat\left(\mathbf{F}_s, \mathbf{F}_f\right)\right)\right) \\
\mathbf{F}_{fs} &= \mathrm{BFT}\left(conv_1\left(cat\left(\mathbf{F}_f, \mathbf{F}_s\right)\right)\right) \\
\mathbf{F}_{sf}^i &= conv_1\left(cat\left(\mathbf{F}_{sf}, \mathbf{F}_{fs}\right)\right). \qquad (9)
\end{aligned}$$

1) Branch Based on Windowed Attention Transformer: The standard self-attention mechanism incurs prohibitively high computational complexity when applied to 2-D images, as both time and memory requirements scale quadratically with the input size. This limitation is especially severe in the context of high-resolution images or hyperspectral data. To enhance efficiency while retaining the capacity to model local structures effectively, we adopt the BWT to capture spatial contextual information. The specific operations of BWT are as follows:

$$\begin{aligned}
\mathbf{F}_s^{mid} &= \mathrm{WSA}\left(LN\left(\mathbf{F}_s^{in}\right)\right) + \mathbf{F}_s^{in} \\
\mathbf{F}_s &= \mathrm{BWT}\left(\mathbf{F}_s^{in}\right) = FFN\left(LN\left(\mathbf{F}_s^{mid}\right)\right) + \mathbf{F}_s^{mid}. \qquad (10)
\end{aligned}$$

Here, $\mathbf{F}_s^{in}$, $\mathbf{F}_s^{mid}$, and $\mathbf{F}_s$, respectively, denote the input feature, the intermediate feature representation, and the output feature of the BWT module. WSA represents the window self-attention module [63], *LN* is layer normalization, and *FFN* denotes the feed-forward network containing two linear layers and a GELU activation function.

2) Branch Based on Frequency-Domain Transformer: Traditional convolution or SA mechanisms are inherently limited to modeling local relationships in the pixel domain, making it difficult to capture information at the frequency level. To address this, the BFT branch maps the input features to the frequency domain and explicitly separates the amplitude and phase information for independent modeling and fusion. We obtain the amplitude **A** and phase **P** through the following formulas:

$$\begin{aligned}
\mathcal{F}\left(\mathbf{I}\right)(u,v) &= \sum_{x=0}^{H-1}\sum_{y=0}^{W-1}\mathbf{I}\left(x,y\right)e^{-2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right)j} \\
\mathbf{A} &= \left|\mathcal{F}\left(\mathbf{I}\right)\right| \\
\mathbf{P} &= \arctan\left(\mathcal{F}\left(\mathbf{I}\right)^{im}/\mathcal{F}\left(\mathbf{I}\right)^{re}\right) \qquad (11)
\end{aligned}$$

where $\mathcal{F}(\cdot)$ denote the 2-D fast Fourier transform (FFT) [64], u and v represent specific horizontal and vertical spatial frequencies in the Fourier spectrum $\mathcal{F}(\mathbf{I})$. The spectrum consists of complex values, expressed as $\mathcal{F}(\mathbf{I}) = \mathcal{F}(\mathbf{I})^{im} + \mathcal{F}(\mathbf{I})^{re} \cdot j$. $\mathcal{F}(\mathbf{I})^{im}$ and $\mathcal{F}(\mathbf{I})^{re}$ are the real and imaginary components, respectively. $|\cdot|$ denotes the modulus operation. **A** and **P** are then separately passed through two convolutional layers with ReLU activation for feature enhancement, resulting in the enhanced amplitude and enhanced phase. The enhanced frequency-domain features are then transformed back to the spatial domain using the inverse fast Fourier transform (IFFT). A Sigmoid function is applied to generate an attention map, which is then multiplied element-wise with the input features that have been enhanced through two convolutions with ReLU activation, resulting in a frequency-domain-based self-attention (FSA) output $\mathbf{F}_f^{out}$. The operation of FSA can be formulated as follows:

$$\mathbf{F}_f^{out} = CRC\left(\mathbf{F}_f^{in}\right)\cdot\left(\sigma\left(\mathcal{F}^{-1}\left(CRC\left(\mathbf{A}\right)e^{CRC(\mathbf{P})j}\right)\right)\right). \qquad (12)$$

Here, *CRC* denotes two convolutions with a 3×3 kernel and a stride of 1, with a ReLU activation function sandwiched between them. $\mathcal{F}^{-1}$ represents the IFFT operator. The operations of BFT can be formulated as follows:

$$\begin{aligned}
\mathbf{F}_f^{mid} &= \mathrm{FSA}\left(LN\left(\mathbf{F}_f^{in}\right)\right) + \mathbf{F}_f^{in} \\
\mathbf{F}_f &= \mathrm{BFT}\left(\mathbf{F}_f^{in}\right) = FFN\left(LN\left(\mathbf{F}_f^{mid}\right)\right) + \mathbf{F}_f^{mid}. \qquad (13)
\end{aligned}$$

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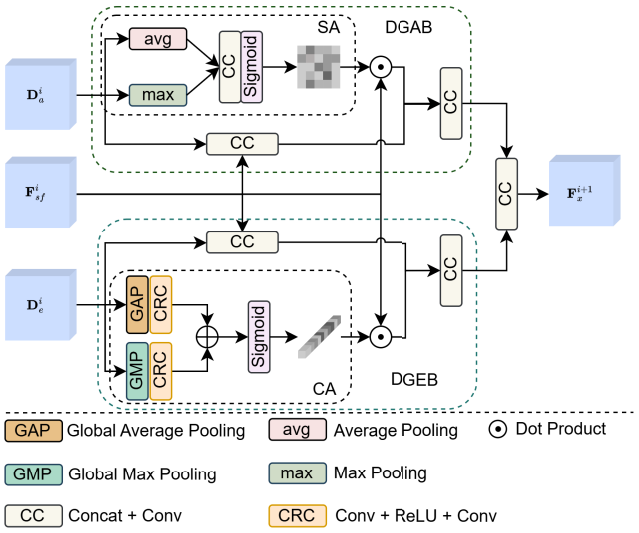


Fig. 4. Architecture of DGM.

E. Degradation Guidance Module

To effectively exploit the degradation information contained in LR-HSI and HR-MSI, we propose DGM, as illustrated in Fig. 4. Specifically, the DGM takes three inputs: 1) the intermediate feature $\mathbf{D}_a^i$, extracted from the LR-HSI by $\boldsymbol{U}_{\text{spa}}$, which carries spatial degradation information; 2) the intermediate feature $\mathbf{D}_e^i$, extracted from the HR-MSI by $\boldsymbol{U}_{\text{spe}}$, which contains spectral degradation information; and 3) the intermediate feature $\mathbf{F}_{sf}^i$, obtained after spatial and frequency-domain restoration through the SFAM, which requires further refinement. The proposed DGM consists of two parallel branches: spatial degradation guidance block (DGAB) and spectral degradation-guided block (DGEB).

DGAB branch can be divided into two parts. In the first part, the input feature $\mathbf{D}_a^i$ is processed through a spatial attention (SA) mechanism [65] to generate a SA map that highlights spatially informative regions. The generated SA map is then applied to the feature $\mathbf{F}_{sf}^i$ requiring refinement, producing a spatially enhanced feature $\mathbf{F}_{ae}^i$, which can be formulated as follows:

$$\mathbf{F}_{ae}^i = \text{SA}\left(\mathbf{D}_a^i\right) \cdot \mathbf{F}_{sf}^i. \tag{14}$$

The second part performs a cascaded operation to fuse $\mathbf{F}_{sf}^i$ and $\mathbf{D}_a^i$, obtaining an intermediate degradation-aware representation $\mathbf{F}_{da}^i$, which is defined as follows:

$$\mathbf{F}_{da}^i = conv_1\left(cat\left(\mathbf{F}_{sf}^i, \mathbf{D}_a^i\right)\right). \tag{15}$$

Finally, the spatially enhanced feature $\mathbf{F}_{ae}^i$ and the intermediate degradation-aware representation $\mathbf{F}_{da}^i$ are concatenated along the channel dimension and fused through a 1×1 convolution to obtain the spatial degradation-guided enhanced feature $\mathbf{F}_{xa}^i$

$$\mathbf{F}_{xa}^i = conv_1\left(cat\left(\mathbf{F}_{ae}^i, \mathbf{F}_{da}^i\right)\right). \tag{16}$$

The DGEB adopts a similar operation as DGAB, except that a channel attention (CA) mechanism [65] is used to generate the CA map, which highlights spectrally informative regions. Finally, the features refined by both spatial and spectral degradation guidance are fused via a 1×1 convolution, which can be formulated as follows:

$$\mathbf{F}_x^{i+1} = conv_1\left(cat\left(\mathbf{F}_{xa}^i, \mathbf{F}_{xe}^i\right)\right) \tag{17}$$

where $\mathbf{F}_{xe}^i$ represents the spectral degradation-guided enhanced feature obtained after processing by the DGEB. $\mathbf{F}_x^{i+1}$ denotes the final feature enhanced by the DGM.

F. Loss Function

The total loss in this study consists of two components and can be expressed as

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha\, \mathcal{L}_{\text{sub}} \tag{18}$$

where $\mathcal{L}_{\text{rec}}$ represents the reconstruction loss, $\mathcal{L}_{\text{sub}}$ is used to constrain the two degradation-aware U-Nets, and α is a weighting factor ($\alpha = 0.5$ in this study). The reconstruction loss $\mathcal{L}_{\text{rec}}$ is used to measure the discrepancy between the reconstructed image and the ground truth, computed as

$$\mathcal{L}_{\text{rec}} = \|\mathbf{X}_r - \mathbf{X}\|_1 \tag{19}$$

where $\|\cdot\|_1$ is the l_1 norm.

Furthermore, to better supervise the two U-Nets ($\boldsymbol{U}_{\text{spa}}$ and $\boldsymbol{U}_{\text{spe}}$) in generating priors for spatial and spectral degradation, we use $\mathcal{L}_{\text{sub}}$ to measure the differences between the reconstructed images and the ground truth

$$\mathcal{L}_{\text{sub}} = \|\mathbf{X}_{\text{ra}} - \mathbf{X}\|_1 + \|\mathbf{X}_{\text{re}} - \mathbf{X}\|_1. \tag{20}$$

We adopt the l_1 loss as the loss metric for both $\mathcal{L}_{\text{rec}}$ and $\mathcal{L}_{\text{sub}}$, as it helps reduce excessive smoothing in the re constructed images. Compared to l_2 loss, l_1 loss better preserves high-frequency details and spectral fidelity by penalizing errors linearly rather than quadratically, which is particularly important for maintaining sharp edges and fine textures in HIF.

IV. Experiments

This section introduces the datasets and preprocessing procedures used in our work, as well as the commonly employed evaluation metrics. In addition, we conduct a series of ablation studies and comparative experiments to demonstrate the effectiveness of the proposed method.

A. Datasets and Experiment Settings

1) Datasets: We conducted experiments on three widely used benchmark datasets to validate the effectiveness of the proposed method: CAVE [66], Harvard [67], and ICVL [68]. The CAVE dataset contains 32 HSIs, each with 31 spectral bands at a spatial resolution of 512×512. The wavelength range spans from 400 to 700 nm with a spectral resolution of 10 nm. In our experiments, 20 images were randomly selected for training, and the remaining 12 images were used for testing.

The Harvard hyperspectral dataset consists of 50 HSIs covering various indoor and outdoor scenes. Each image has 31 narrow spectral channels, with wavelengths ranging from 420 to 720 nm at 10 nm intervals, and a spatial resolution of 1392×1040. In our experiments, we cropped the top-left

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corner of each image to obtain a size of 1024 × 1024. We randomly selected 30 images for training and the remaining 20 images for testing.

The ICVL hyperspectral dataset, collected and released by the Computational Vision Laboratory at Ben-Gurion University, contains 200 HSIs. Each image has 31 spectral bands with a wavelength range of 400–700 nm, a spectral interval of 10 nm, and a spatial resolution of 1392 × 1300. For our experiments, we cropped the top-left corner of each image to a size of 1280 × 1280. Similar to Harvard, 20 images were randomly selected for training and 10 images for testing.

2) Details of PSF and SRF: Regarding PSFs, we recognize that spatial degradation in real remote sensing systems exhibits diverse characteristics. To capture this diversity while maintaining physical interpretability, we generate two categories of PSFs based on well-established optical models: five isotropic Gaussian kernels and 30 anisotropic Gaussian kernels, all with a fixed kernel size of 15 × 15. The isotropic Gaussian kernels simulate optical blur caused by diffraction and defocus effects, and the calculation formula is

$$G_{\text{iso}}(x,y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{21}$$

where (x, y) represents the coordinates of each pixel within the Gaussian kernel relative to the kernel center. The standard deviation is given by $\sigma = (\mu r^2/8\ln(2))$, where r denotes the downsampling scale and μ is the parameter taking values from the set $\{0.3, 0.4, 0.5, 0.6, 0.7\}$.

The anisotropic Gaussian kernels simulate more complex degradation patterns, including motion blur and sensor integration effects commonly encountered in push-broom scanning systems. The calculation formula is

$$\begin{aligned} G_{\text{aniso}}(x,y) &= \frac{1}{2\pi\sigma_x\sigma_y}\exp\left(-\frac{x'^2}{2\sigma_x^2}-\frac{y'^2}{2\sigma_y^2}\right) \\ x' &= x\cos\omega + y\sin\omega \\ y' &= -x\sin\omega + y\cos\omega. \end{aligned} \tag{22}$$

Here, ω represents the principal axis direction of the Gaussian kernel, with values taken from $\{0, \pi/12, \pi/6, \pi/4, \pi/3\}$; and (σ_x, σ_y) denote the standard deviations along the principal axis directions, consisting of six scale combinations $\{\{1,5\},\{1,9\},\{3,7\},\{5,9\},\{7,1\},\{7,5\}\}$. By combining the directional parameters with the scale parameters, a total of 30 anisotropic kernels are generated. Together with the aforementioned five isotropic Gaussian kernels, we construct 35 physically meaningful PSFs designed to span spatial degradation types ranging from mild to severe and from isotropic to anisotropic. These parameter ranges are consistent with the degradation characteristics observed in real remote sensing sensors.

For SRFs, we adopt 28 real camera spectral sensitivity functions from the publicly available camera spectral sensitivity database [69], which includes professional digital single-lens reflex, point-and-shoot, industrial, and mobile phone cameras, with measurements ranging from 400 to 720 nm at 10 nm intervals.

These PSFs and SRFs are partitioned into two nonoverlapping subsets, where the training set comprises 27 PSFs and 20 SRFs, and the test set contains 8 PSFs and 8 SRFs. Training and testing data are synthesized by randomly sampling PSFs and SRFs from separate subsets, respectively, with details illustrated in Figs. 5 and 6.

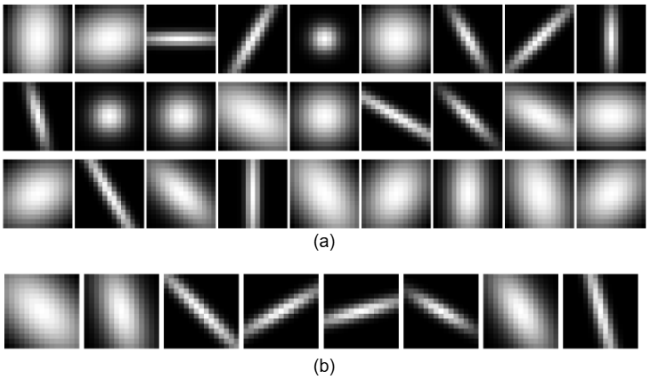


Fig. 5. Illustration of the adopted PSFs. (a) PSFs for training. (b) PSFs for inference.

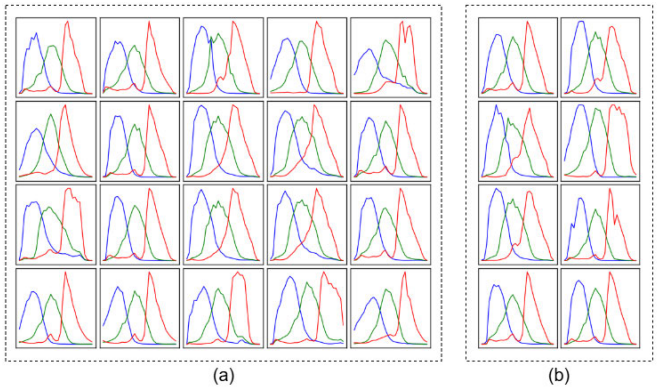


Fig. 6. Illustration of the adopted SRFs. (a) SRFs for training. (b) SRFs for inference.

3) Experimental Data Processing: We followed the Wald protocol [70] to generate the observed images for the simulated datasets: LR-HSIs and HR-MSIs. To generate LR-HSIs, we randomly selected a PSF from the corresponding subset for each image to blur the reference image. The blurred image was then downsampled by a factor of 8, with the downsampling method randomly chosen from four strategies, i.e., {"*stride sampling*," "*bicubic interpolation*," "*linear interpolation*," "*nearest-neighbor interpolation*"}. To generate HR-MSIs, we randomly selected an SRF from the corresponding subset for spectral downsampling. To simulate ubiquitous noise, Gaussian noise of moderate intensity was added to LR-HSIs (SNR = 30 dB) and HR-MSIs (SNR = 40 dB). In our experiments, image patches of size 64 × 64 and 8 × 8 were cropped from the observed LR-HSIs and HR-MSIs, respectively, to serve as inputs. During training, we employ a sliding window sampling strategy with dataset-specific strides to increase training sample diversity: stride of 16 pixels for the CAVE dataset, and stride of 48 pixels for both the Harvard and ICVL datasets. This sampling strategy generates a total of 16 820 patches from the CAVE dataset, 13 230 patches

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from the Harvard dataset, and 13 520 patches from the ICVL dataset. For testing, we use a sliding window strategy with a stride of 32 pixels to crop each test image into 64 × 64 patches.

4) Quantitative Assessment Metrics: We evaluate the fusion results from both visual and quantitative perspectives. Four commonly used metrics are employed to assess the performance of different methods: peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [71], structural similarity index measure (SSIM) [72], and erreur relative globale adimensionnelle de synthése (ERGAS) [73]. PSNR measures the ratio between the maximum signal intensity and the noise in an image, reflecting the error between the reconstructed image and the reference image. Higher PSNR values generally indicate lower noise and less distortion, implying better reconstruction quality. SAM evaluates the similarity between the reconstructed and reference spectra by computing the angle between two spectral vectors. Smaller SAM values indicate smaller spectral angle differences, suggesting that the reconstructed spectra are more consistent with the ground truth. SSIM assesses the similarity between the reconstructed and reference images in terms of structure, luminance, and contrast. Higher SSIM values indicate better preservation of image details and overall visual consistency. ERGAS quantifies the global deviation of the reconstructed image at both spectral and spatial levels. It is calculated based on the ratio of the root mean square error to the mean across all bands, providing a comprehensive measure of the overall reconstruction distortion. Smaller ERGAS values indicate higher consistency with the reference image in terms of spectral and spatial information. In addition, we provide visualizations of the reconstructed images and residual maps with respect to the ground truth to intuitively illustrate the reconstruction quality.

5) Comparison Methods: To comprehensively evaluate the effectiveness of the proposed method, we selected eight representative methods for comparison, including CNMF [27], EDBIN [74], PSRT [75], GuidedNet [38], MIMO-SST [43], DSPNet [76], PSTUN [77], and ASMNet [78]. To ensure the reliability of the results, all experiments were conducted using the same datasets for training and testing. For the traditional method CNMF, experiments were implemented in MATLAB 2019a. For the remaining deep learning-based methods, experiments were conducted in the PyTorch framework on a GeForce GTX 3090Ti GPU with 24 GB memory. During training, the Adam optimizer was employed for 500 epochs, with an initial learning rate of 5×10^{-4}, which was decayed to half of the original value every 100 epochs. For fair comparison, all deep learning-based methods were implemented using official code or reproduced strictly following the original papers. All models were optimized using the Adam optimizer. Specifically, EDBIN, PSRT, MIMO-SST, PSTUN, and ASMNet were trained for 500 epochs, while GuidedNet and DSPnet were trained for 150 epochs. The learning rates were set to 1×10^{-3} for EDBIN, 1×10^{-4} for PSRT, MIMO-SST and ASMNet, 1×10^{-5} for GuidedNet, 3×10^{-4} for DSPNet, and 2×10^{-4} for PSTUN. Batch sizes were 32 for EDBIN, PSRT and GuidedNet, 16 for MIMO-SST and ASMNet, 24 for DSPNet, and 20 for PSTUN.

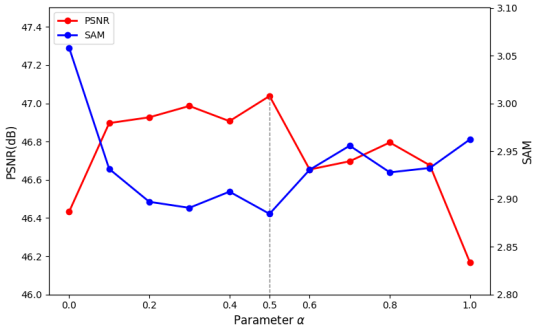


Fig. 7. PSNR and SAM values as measures on parameter α.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT METHODS TRAINED ON SINGLE-DEGRADATION(S) AND MULTI-DEGRADATION(M) ON HARVARD DATASET. THE BEST RESULTS ARE IN BOLD

Methods	Training	PSNR↑	SAM↓	SSIM↑	ERGAS↓
EDBIN	S	43.4088	3.3660	0.9861	1.1918
	M	44.5281	3.1990	0.9865	1.0506
PSRT	S	42.3175	2.9419	0.9830	1.1425
	M	43.7742	3.5281	0.9848	1.1218
GuidedNet	S	43.6267	3.4264	0.9854	1.2103
	M	45.6272	3.3474	0.9849	1.1111
MIMO-SST	S	43.7016	3.3313	0.9829	1.1711
	M	45.6961	3.1199	0.9863	0.9496
DSPNet	S	44.5996	3.1073	0.9866	1.0514
	M	46.3306	3.0093	0.9873	0.9323
PSTUN	S	43.3745	3.2973	0.9863	1.1993
	M	45.3166	3.0943	0.9872	0.9941
ASMNet	S	43.6151	3.3541	0.9859	1.1938
	M	45.4903	3.0702	0.9867	0.9868
Ours	S	45.4062	3.1018	0.9868	0.9852
	M	**47.0384**	**2.8844**	**0.9878**	**0.8469**

B. Ablation and Parameter Analysis

In our study, we conducted ablation experiments on the Harvard dataset. To validate the hypothesis that simulating diverse degradation conditions enables the network to learn latent causal relationships independent of specific degradation mechanisms, we generated new training data using a single degradation type. We then retrained all deep learning methods and evaluated them on multiple unseen degradation scenarios, with the results summarized in Table I. The results show that when trained with only a single degradation, the performance of all deep learning methods drops noticeably under unseen degradations. For example, the PSNR of ASMNet decreases from 45.49 to 43.61 dB, and SAM increases from 3.07 to 3.35. For our method, PSNR decreases from 47.04 dB to 45.40 dB, and SAM rises from 2.88 to 3.10, indicating a strong dependence on specific degradation mechanisms. These results validate the effectiveness of the multidegradation training strategy: by exposing the network to a variety of degradation conditions, it is forced to learn the core causal mapping rather than merely fitting specific degradation patterns, thereby achieving robust performance on unseen degradations.

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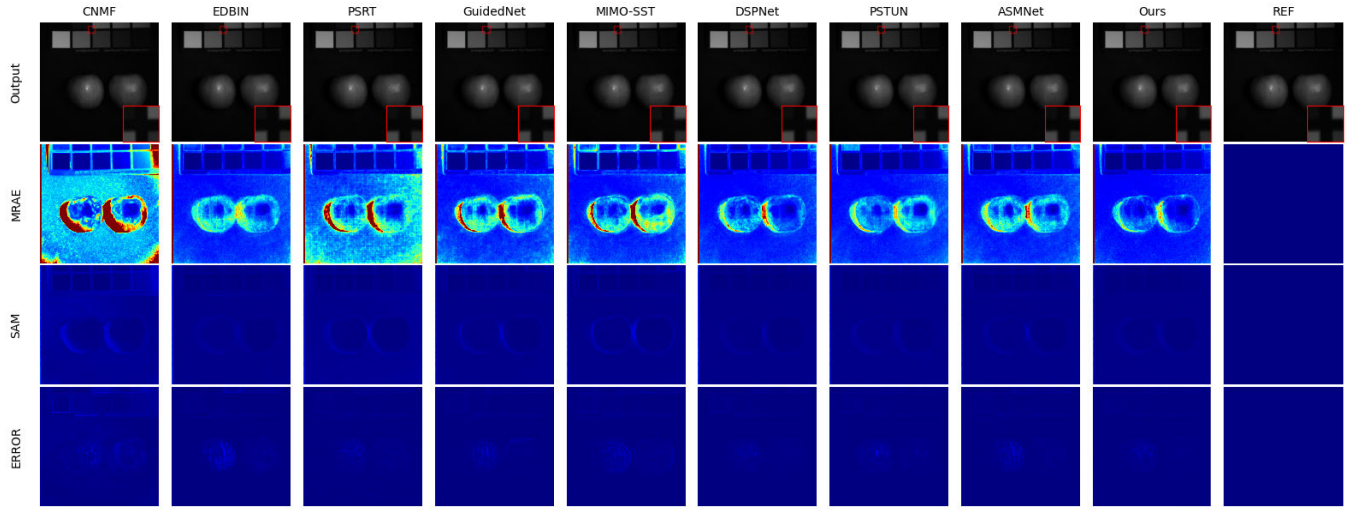


Fig. 8. Fusion results in the CAVE dataset (output and error map correspond to band 29).

TABLE II

QUANTITATIVE EVALUATION OF ABLATION EXPERIMENTS ON HARVARD DATASET. THE BEST RESULTS ARE IN BOLD

DGAB	DGEB	SFAM	PSNR↑	SAM↓	SSIM↑	ERGAS↓
×	✓	✓	46.0733	2.9714	0.9828	0.9219
✓	×	✓	46.7572	2.9381	0.9875	0.8759
✓	✓	×	46.6931	3.0024	0.9872	0.8929
×	×	✓	45.8634	3.0541	0.9868	0.9467
✓	✓	✓	**47.0384**	**2.8844**	**0.9878**	**0.8469**

TABLE III

ABLATION STUDY ON SFAM COMPONENTS: EVALUATION OF SPATIAL BRANCH, FREQUENCY BRANCH, AND BIDIRECTIONAL INTERACTION ON HARVARD DATASET. THE BEST RESULTS ARE IN BOLD

Spatial	Frequency	Interaction	PSNR↑	SAM↓	SSIM↑	ERGAS↓
×	×	×	46.6931	3.0024	0.9872	0.8929
✓	×	×	46.8755	2.9457	**0.9879**	0.8927
×	✓	×	47.0235	2.9215	**0.9879**	0.8802
✓	✓	×	47.0121	2.9726	0.9878	0.8810
✓	✓	✓	**47.0384**	**2.8844**	0.9878	**0.8469**

In addition, we also validated the effectiveness of the proposed deep learning network components, including DGAB, DGEB, and SFAM, with the experimental results summarized in Table II. By removing each component from the network, we evaluated their individual contributions. Specifically, when the DGAB module is absent, PSNR drops from 47.04 to 46.07 dB and ERGAS rises from 0.84 to 0.92, indicating that DGAB plays a critical role in enhancing image details and preserving global structure. DGEB is essential for suppressing spectral degradation errors and improving spectral fidelity. When SFAM is removed, SAM increases significantly (2.88–3.00) while PSNR slightly decreases. To further validate the effectiveness of SFAM's internal design, we conducted ablation studies on its variants: using only the spatial branch, using only the frequency branch, and using both branches with simple serial concatenation. The results are reported in Table III. Compared to using individual branches or simple serial concatenation, the full SFAM with bidirectional interaction achieves the best performance (PSNR 47.04 dB, SAM 2.88), demonstrating that the complete SFAM is indispensable for integrating spectral and spatial information while maintaining spectral fidelity. However, when these components are combined, their effects are not merely additive but exhibit nonlinear synergistic interactions. Through this cooperation, the modules collectively achieve overall performance optimization.

In the model parameter sensitivity analysis, we investigated the effect of the parameter α in the loss function used during training. Fig. 7 illustrates the sensitivity of the fusion performance to the balance parameter α in the total loss function. A small α (e.g., 0.1) under-weights the subnetworks loss $\mathcal{L}_{\text{sub}}$, providing insufficient supervision to $\boldsymbol{U}_{\text{spa}}$ and $\boldsymbol{U}_{\text{spe}}$, which degrades spectral fidelity and increases SAM. A large α (e.g., 0.9) over-emphasizes $\mathcal{L}_{\text{sub}}$ at the cost of the reconstruction loss $\mathcal{L}_{\text{rec}}$, reducing spatial detail injection and lowering PSNR. The optimal setting $\alpha = 0.5$ achieves the best balance, yielding the highest PSNR and lowest SAM.

C. Comparison Results of Experiments on Simulated Datasets

1) Results on CAVE Dataset: Table IV presents a comparison of the experimental results of different methods on the CAVE dataset, with the best results highlighted in bold. Our method outperforms all other approaches across all evaluation metrics, achieving the highest PSNR and SSIM as well as the lowest SAM and ERGAS. Among the deep learning baselines, GuidedNet and ASMNet show comparable performance, while DSPNet exhibits relatively stronger spatial–spectral preservation. The traditional method CNMF performs significantly worse, highlighting its limitations in handling high-dimensional spectral information. To provide a more intuitive comparison of the fusion performance of each method, we present MRAE and SAM heatmaps as well as error maps for the *yellowpeppers* image in Fig. 8. MRAE reflects the overall quality of the fused image, SAM measures

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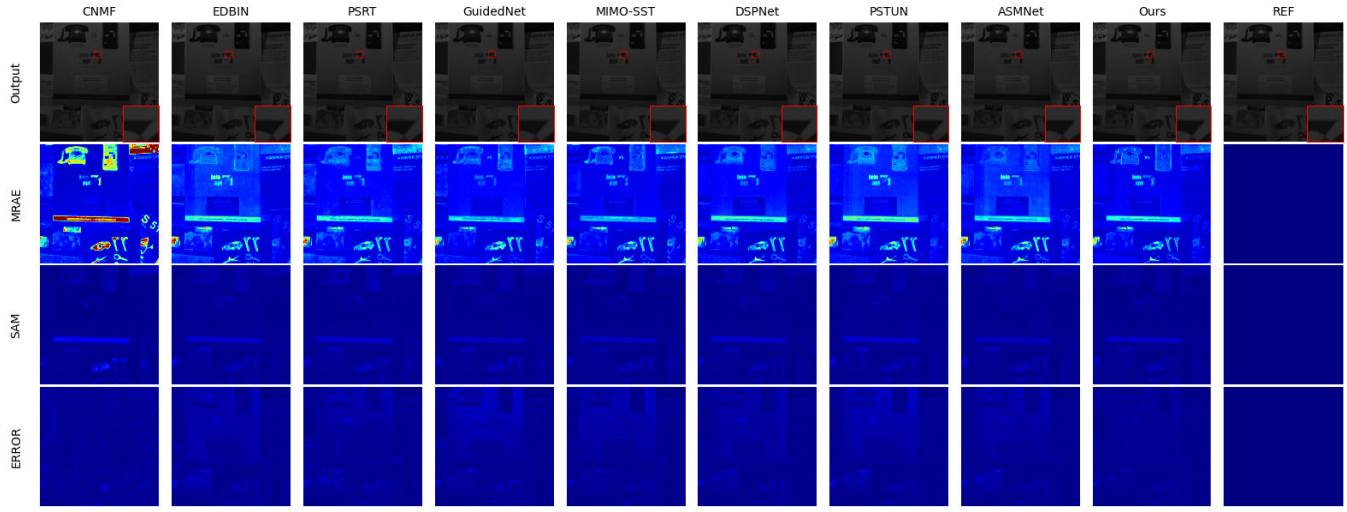


Fig. 9. Fusion results in the Harvard dataset (output and error map correspond to band 9).

spectral deviations, and the error maps illustrate the closeness of specific bands in the fused image to the reference image. These visualizations show that our method has the smallest differences from the reference image, clearly outperforming the other approaches. Overall, our method better preserves both spatial and spectral information on the CAVE dataset, resulting in higher fusion quality. To further highlight performance differences among the methods, we computed the PSNR across all spectral bands, as shown in Fig. 11(a). It can be observed that our method consistently outperforms others across most bands, demonstrating its excellent adaptability and robustness. In contrast, CNMF exhibits significant drops in all metrics, with higher errors and notably lower PSNR values, reflecting its limited generalization capability.

2) Results on Harvard Dataset: Table V presents the experimental results of all compared methods on the Harvard dataset. In comparison, our method achieves the best performance across spatial fidelity metrics such as PSNR and SSIM, as well as spectral fidelity metrics such as SAM and ERGAS, followed by DSPNet and MIMO-SST, demonstrating its clear advantage in fusing spatial and spectral information. To provide a more intuitive comparison of the fusion results, we visualized the image *im1_02* from the Harvard dataset, as shown in Fig. 9. It can be observed that CNMF, EDBIN, PSRT, and GuidedNet exhibit noticeable distortions, whereas our method performs best in both detail restoration and spectral fidelity, with minimal differences from the reference image. This further indicates that our network can effectively capture the complex relationships between spatial and spectral information, achieving high-quality image reconstruction. Fig. 11(b) presents the PSNR values across all spectral bands, further validating the quantitative results. Our network maintains a leading advantage across most bands, demonstrating strong robustness and adaptability. Overall, the experimental results on the Harvard dataset indicate that our method outperforms existing approaches both visually and across most spectral bands, fully demonstrating its effectiveness and reliability in HIF tasks.

TABLE IV
QUALITY METRICS OF DIFFERENT METHODS IN CAVE DATASET. THE BEST RESULTS ARE IN BOLD

Method	PSNR↑	SAM↓	SSIM↑	ERGAS↓
Best Value	$+\infty$	0	1	0
CNMF	39.8893	8.5469	0.9624	1.4568
EDBIN	42.4043	3.9938	0.9886	1.0775
PSRT	43.3402	5.0869	0.9840	1.0535
GuidedNet	44.0473	4.1337	0.9877	0.9518
MIMO-SST	43.3573	4.3090	0.9864	0.9530
DSPNet	44.9621	3.5566	0.9905	0.8492
PSTUN	42.1095	4.0242	0.9881	1.1994
ASMNet	44.1992	3.8295	0.9890	0.8933
Ours	**45.2206**	**3.3795**	**0.9908**	**0.8395**

3) Results on ICVL Dataset: Table VI presents the experimental results of all compared methods on the ICVL dataset. It can be observed that our method achieves the best performance in both spatial fidelity metrics (PSNR and SSIM) and spectral fidelity metrics (SAM and ERGAS), followed by DSPNet, further demonstrating the strong generalization capability of our approach across different scenarios. Fig. 10 shows the visualized fusion results of the *objects_0924-1625* image from the ICVL dataset for all methods. The results indicate that our method performs best across these visualizations, with the smallest errors, producing fused images that are highly consistent with the reference images and clearly outperforming other methods in terms of detail recovery and spectral accuracy. In addition, Fig. 11(c) presents the PSNR values across all spectral bands for each method. It can be seen that our method maintains a leading advantage across the majority of bands, reflecting its robustness and adaptability to different spectral regions. Overall, the experimental results on the ICVL dataset demonstrate that our method outperforms existing approaches in spatial detail preservation, spectral fidelity, and full-band performance, fully validating the stability and generalization capability of the model.

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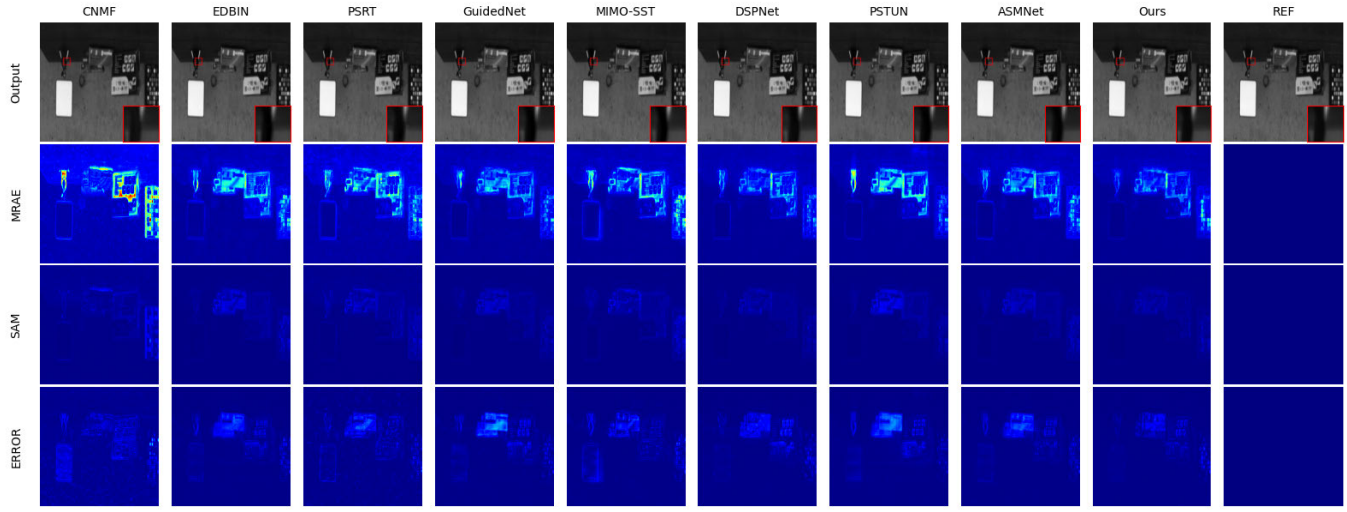


Fig. 10. Fusion results in the ICVL dataset (output and error map correspond to band 9).

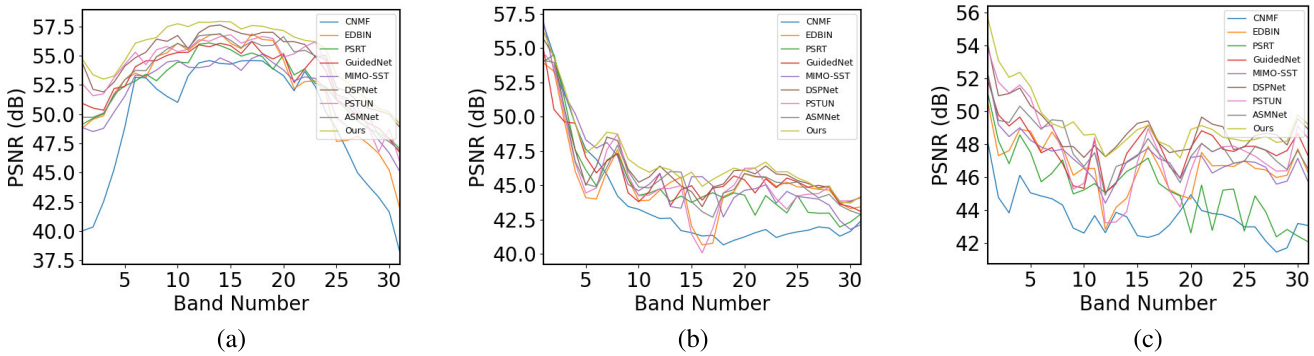


Fig. 11. PSNR values of different methods on three datasets. (a) CAVE. (b) Harvard. (c) ICVL.

TABLE V

QUALITY METRICS OF DIFFERENT METHODS IN HARVARD DATASET. THE BEST RESULTS ARE IN BOLD

Method	PSNR↑	SAM↓	SSIM↑	ERGAS↓
Best Value	$+\infty$	0	1	0
CNMF	43.7576	3.8618	0.9773	1.1326
EDBIN	44.5281	3.1990	0.9865	1.0506
PSRT	43.7742	3.5281	0.9848	1.1218
GuidedNet	45.6272	3.3474	0.9849	1.1111
MIMO-SST	45.6961	3.1199	0.9863	0.9496
DSPNet	46.3306	3.0093	0.9873	0.9323
PSTUN	45.3166	3.0943	0.9872	0.9941
ASMNet	45.4903	3.0702	0.9867	0.9868
Ours	**47.0384**	**2.8844**	**0.9878**	**0.8469**

TABLE VI

QUALITY METRICS OF DIFFERENT METHODS IN ICVL DATASET. THE BEST RESULTS ARE IN BOLD

Method	PSNR↑	SAM↓	SSIM↑	ERGAS↓
Best Value	$+\infty$	0	1	0
CNMF	47.1512	2.2941	0.9875	0.5123
EDBIN	48.7034	1.3788	0.9959	0.3790
PSRT	47.6103	1.9478	0.9939	0.4211
GuidedNet	50.4530	1.3867	0.9959	0.3541
MIMO-SST	49.4628	1.4433	0.9954	0.3387
DSPNet	52.0399	1.2028	0.9964	0.2694
PSTUN	49.5167	1.2786	0.9963	0.3700
ASMNet	50.1738	1.3101	0.9960	0.3376
Ours	**52.3154**	**1.1135**	**0.9967**	**0.2465**

D. Experiment Results on Real Data

We conducted further experiments on the University of Houston (UH) dataset [79] to demonstrate the performance of our method in practical applications. The UH dataset, released by the 2018 IEEE GRSS Fusion Contest, covers the UH and its surrounding areas. It includes a LR-HSI of size $4172 \times 1202 \times 48$ and a corresponding HR-MSI of size $83,440 \times 24,040 \times 3$. For simplicity, we selected a $1024 \times 1024 \times 48$ region from the LR-HSI as the observation data and resampled the corresponding HR-MSI region to $8192 \times 8192 \times 3$, resulting in a resolution ratio of 8. The selected LR-HSI was used as the reference image and spatially downsampled by a factor of 8 to obtain the training LR-HSI. Similarly, the HR-MSI was spatially downsampled for training purposes. Finally, overlapping patches of size $8 \times 8 \times 48$ and $64 \times 64 \times 3$ were extracted for training.

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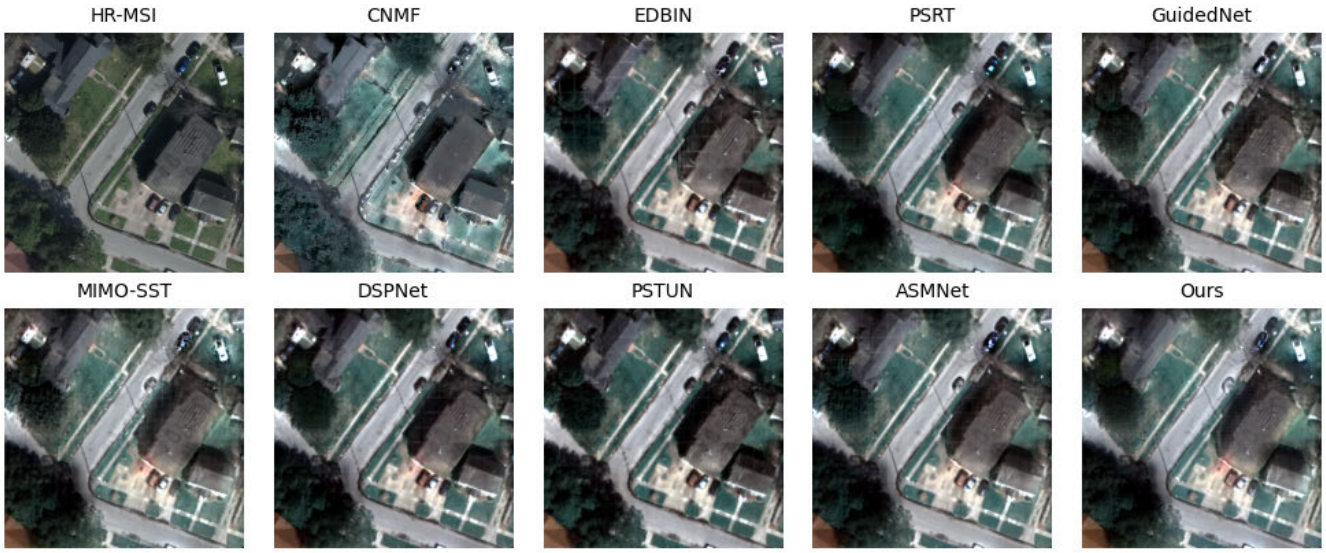


Fig. 12. Visualization of results for the UH dataset (the R-G-B channels correspond to band 18–13–10).

TABLE VII
QUALITY MEASURES WITHOUT MEFERENCE FOR THE UH DATASET USING DIFFERENT METHODS. THE BEST RESULTS ARE IN BOLD

Methods	$D_\lambda \downarrow$	$D_s \downarrow$	QNR↑
Best Value	0	0	1
CNMF	0.0760	0.1602	0.7758
EDBIN	0.0439	0.0857	0.8740
PSRT	0.0340	0.0847	0.8840
GuidedNet	0.0391	0.0863	0.8778
MIMO-SST	0.0459	0.0822	0.8755
DSPNet	0.0298	0.0946	0.8798
PSTUN	0.0347	0.1030	0.8658
ASMNet	0.0302	0.0730	0.8989
Ours	**0.0293**	**0.0646**	**0.9078**

Fig. 12 presents the visualization results on the UH dataset. As shown, our method produces noticeably sharper spatial details. In the absence of ground-truth HR-HSI, we evaluate the fusion performance using several no-reference quality metrics, including the spectral distortion D_λ, the spatial distortion D_s, and the overall quality index QNR [80]. Table VII summarizes the performance of various methods on these metrics. Our approach achieves the best results across all three indicators, further demonstrating the effectiveness and superiority of the proposed method.

E. Computational Efficiency

To provide a comprehensive comparison, it is necessary to analyze the efficiency and computational cost of deep learning-based methods. Table VIII presents the comparison of different methods in terms of model parameters, computational complexity (FLOPs), and test time. For computational efficiency analysis, all measurements were conducted on the same NVIDIA GeForce RTX 3090Ti GPU with a batch size of 1. The reported inference times represent the average processing time per test image on the Harvard dataset. It can be observed that MIMO-SST and DSPNet have relatively large parameter counts and FLOPs, but MIMO-SST exhibits a relatively shorter test time, indicating a certain efficiency advantage during inference. In contrast, PSRT has the smallest number of parameters and relatively low computational cost, yet its test time is not the shortest, which may be influenced by the complexity of its network architecture. GuidedNet, ASMNet, and PSTUN have moderate parameter counts and FLOPs, and their test times are relatively balanced. Our model lies in the middle in terms of both parameters and FLOPs (2.18M parameters and 2.92G FLOPs), while achieving a fast test time of 3.77 s, demonstrating good computational efficiency and practical applicability while maintaining high performance. Overall, these results indicate that our method achieves a well-balanced tradeoff between performance and efficiency, effectively considering model size, computational cost, and inference speed.

TABLE VIII
COMPARISON OF PARAMETERS, FLOPS, AND TEST TIME OF DIFFERENT METHODS

Methods	**PARAMs**	**FLOPs**	**TIME**
EDBIN	889.77K	12.86G	9.28s
PSRT	247.46K	1.04G	6.50s
GuidedNet	1.45M	1.76G	3.81s
MIMO-SST	4.98M	1.56G	2.23s
DSPNet	6.06M	6.59G	5.56s
PSTUN	1.81M	1.48G	3.85s
ASMNet	1.25M	1.22G	4.04s
Ours	2.18M	2.92G	3.77s

V. CONCLUSION

This work revisits the HIF task from a causal perspective. To enhance the model's generalization under unseen degradations, we propose generating diverse training data by randomly adopting multiple PSFs and SRFs, encouraging the network to learn latent causal patterns independent of specific degradation mechanisms. To this end, we develop a degradation-guided blind HIF network equipped with a

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spatial–frequency attention mechanism. Two U-Net subnetworks extract spatial–spectral degradation features, which are then embedded into the reconstruction network to achieve more accurate HR-HSI reconstruction. The SFAM further exploits the complementary information between spatial and frequency domains, improving both spatial detail recovery and spectral consistency.

Although experimental results demonstrate the effectiveness of the proposed method, certain limitations remain. For instance, we attempt to describe unknown degradations using combinations of a given degradation set, yet the completeness of this set still requires further improvement. Future work will focus on further exploring the causal structures in HSIs and integrating causal representation learning with multimodal information. This is expected to enhance the model's robustness and generalization in real-world scenarios, enabling high-quality HR-HSI reconstruction under diverse unseen degradations.

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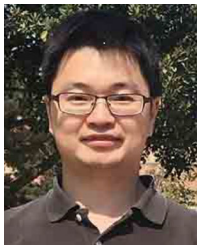
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